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EXPERIMENT-1

AIM: To plot the basis signals

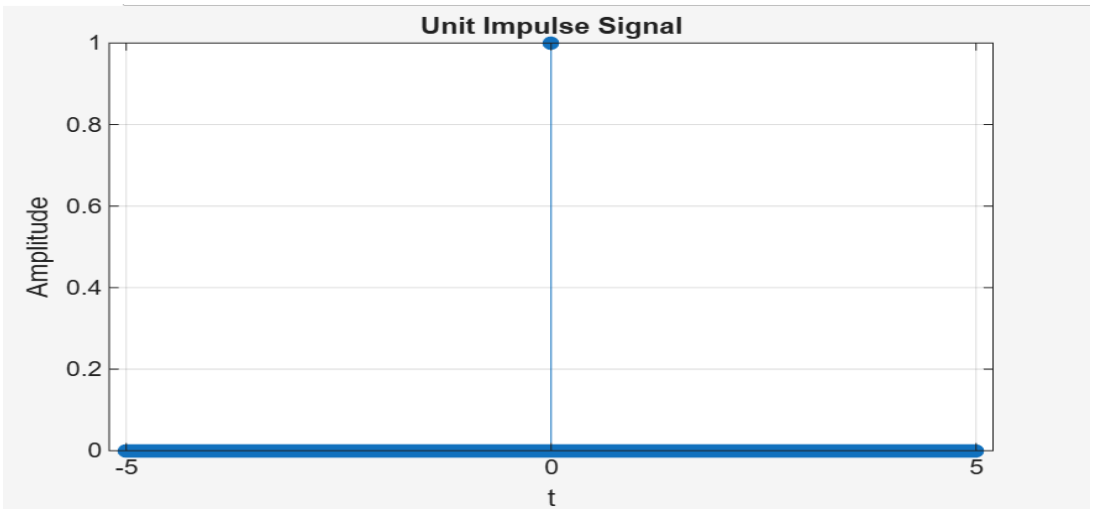
1. Unit impulse 2. Unit step 3.Ramp unit 4. Exponential signals 5. Sine 6. Cos

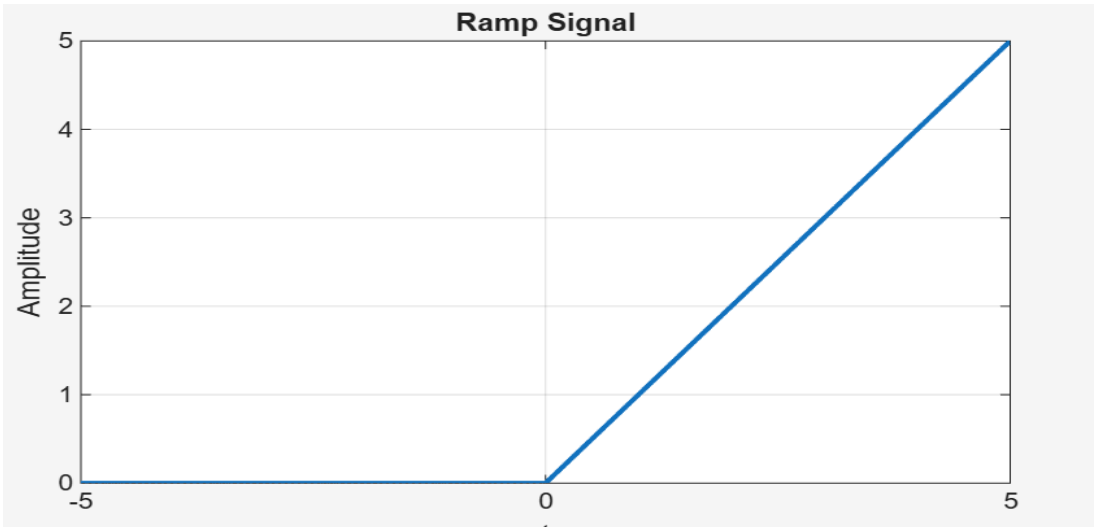
```
clc;
clear;
close all;
% Time axis
t = -5:0.01:5;

1. Unit Impulse
impulse = zeros(size(t));
[~, idx] = min(abs(t));
impulse(idx) = 1;

 2. Unit Step
u = zeros(size(t));
u(t >= 0) = 1;

 3. Ramp Signal
r = zeros(size(t));
r(t >= 0) = t(t >= 0);

 4. Exponential Signal
a = 1; % change for growth/decay
exp_signal = exp(a*t);

 5. Sine Signal
sin_signal = sin(t);

 6. Cosine Signal
cos_signal = cos(t);
```

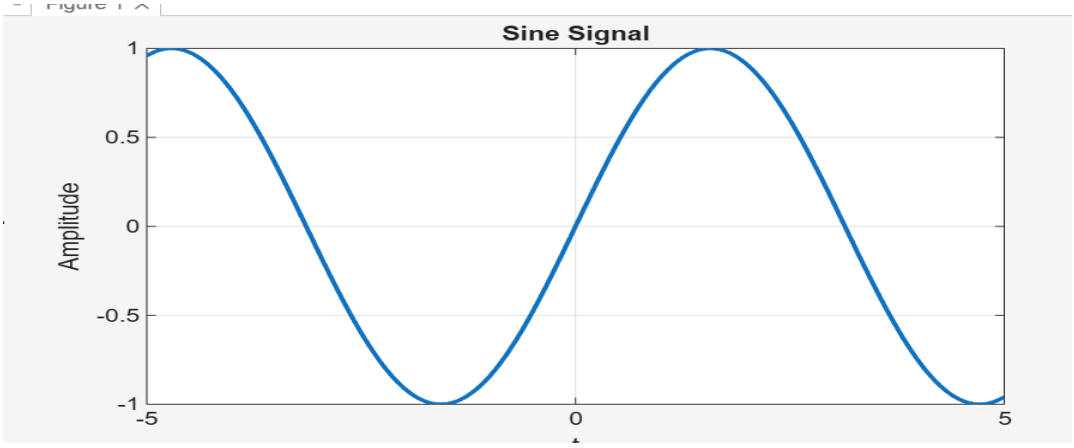
Plotting all signals

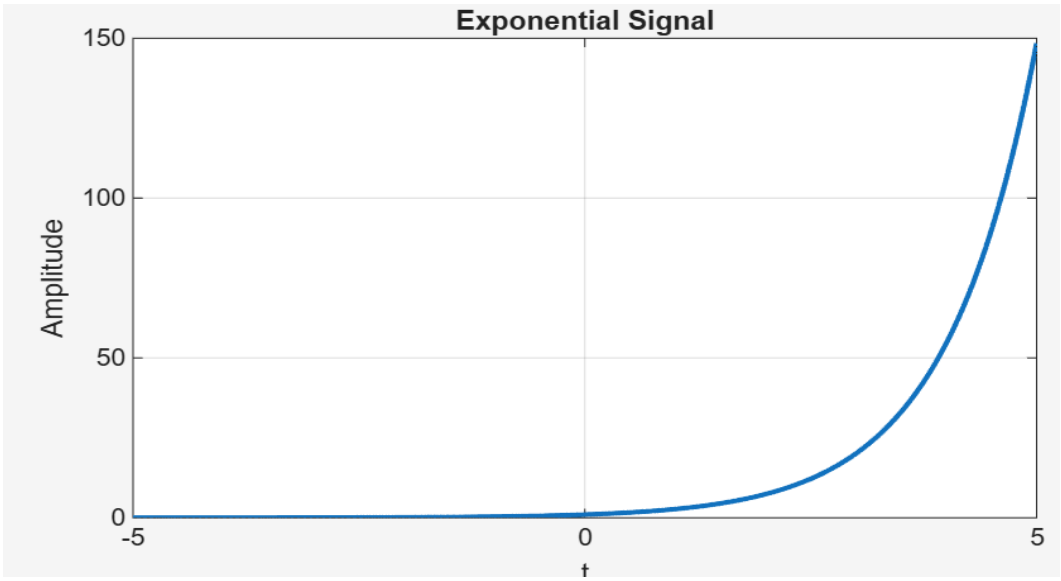
```
figure;

subplot(3,2,1);
stem(t, impulse, 'filled');
title('Unit Impulse Signal');
xlabel('t'); ylabel('Amplitude');
grid on;

subplot(3,2,2);
plot(t, u, 'LineWidth', 2);
title('Unit Step Signal');
xlabel('t'); ylabel('Amplitude');
grid on;

subplot(3,2,3);
plot(t, r, 'LineWidth', 2);
title('Ramp Signal');
xlabel('t'); ylabel('Amplitude');
grid on;

subplot(3,2,4);
plot(t, exp_signal, 'LineWidth', 2);
title('Exponential Signal');
xlabel('t'); ylabel('Amplitude');
grid on;

subplot(3,2,5);
plot(t, sin_signal, 'LineWidth', 2);
title('Sine Signal');
xlabel('t'); ylabel('Amplitude');
grid on;

subplot(3,2,6);
plot(t, cos_signal, 'LineWidth', 2);
title('Cosine Signal');
```

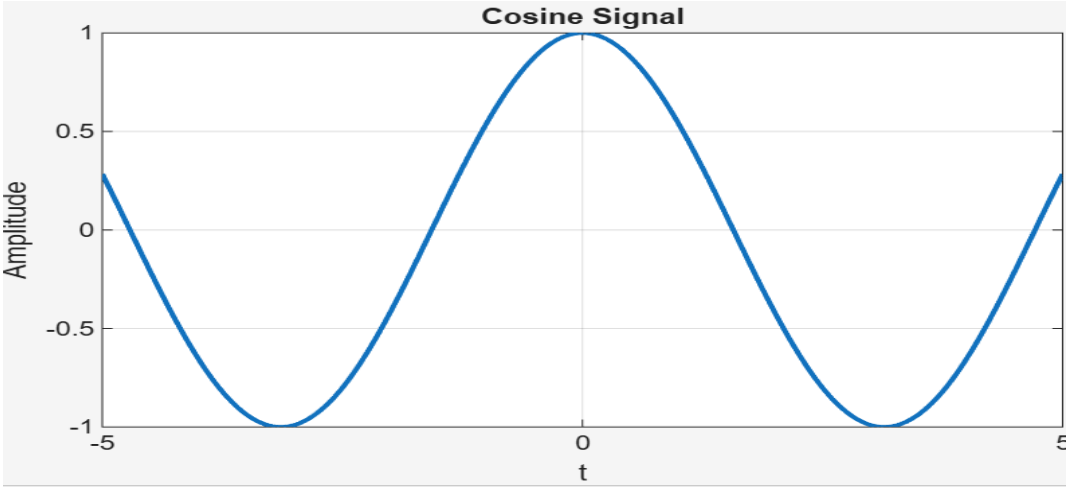
xlabel('t'); ylabel('Amplitude');
grid on;








Exponential Signal
150
100
50
0
Amplitude
-5
0
5
t

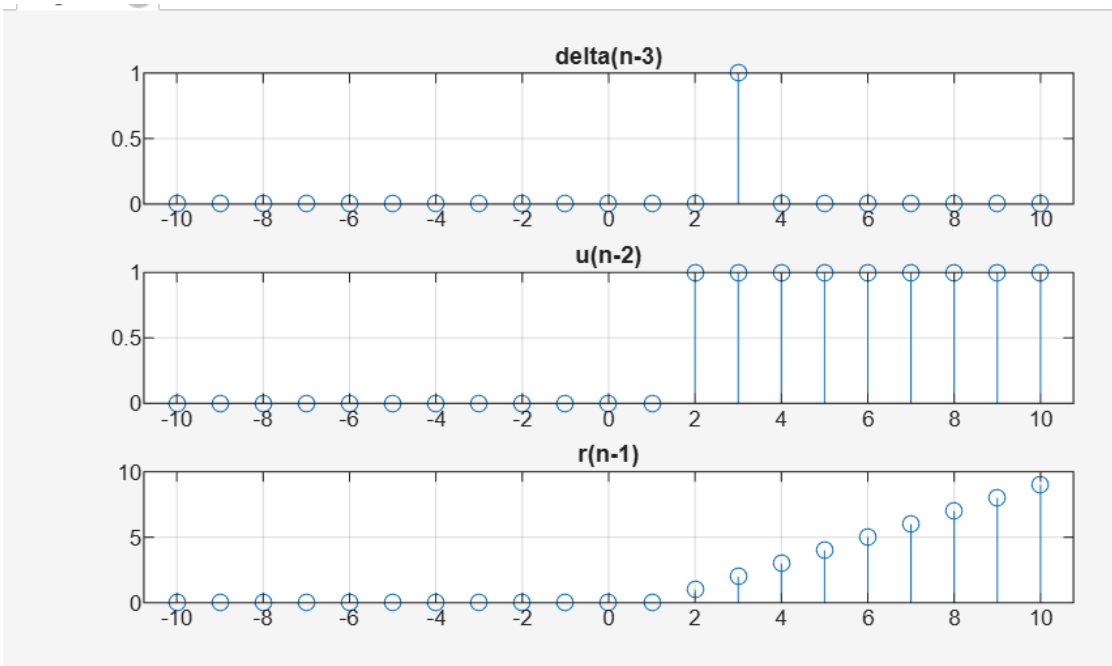

Cosine Signal
1
0.5
0
-0.5
-1
Amplitude
-5
0
5
t

EXPERIMENT-2

AIM: Generate (n-3),u(n-2),r(n-1)

Code

```
clc
clear
close all
n = -10:10;
% Impulse δ(n-3)
impulse = (n == 3);
figure
subplot(3,1,1)
stem(n, impulse)
title('delta(n-3)')
grid on
% Step u(n-2)
u = (n >= 2);
subplot(3,1,2)
stem(n, u)
title('u(n-2)')
grid on
% Ramp r(n-1)
r = n - 1;
r(n < 1) = 0;
subplot(3,1,3)
stem(n, r)
title('r(n-1)')
grid on
```



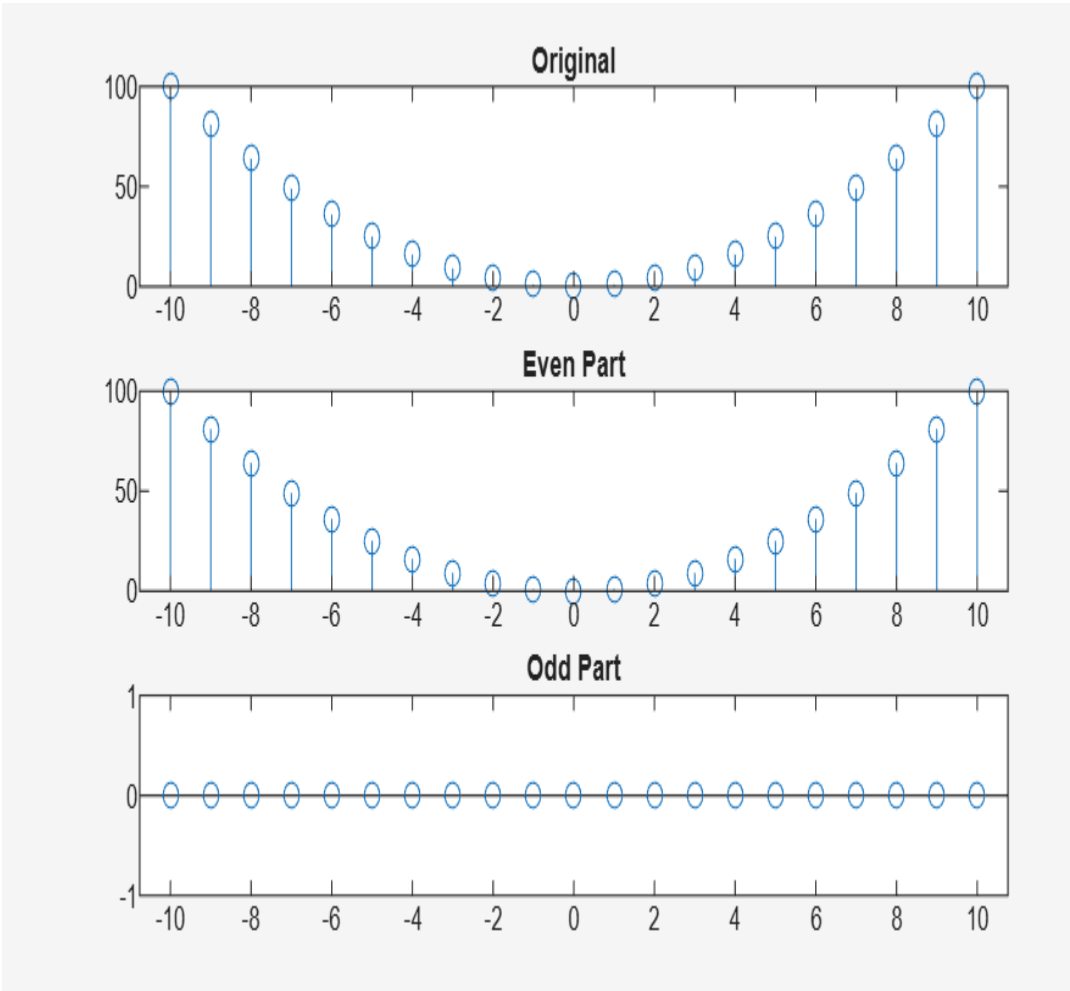
b) Given Signal x(n) = 3u(n) - 2u(n-4)

```
clc;
clear;
close all;
n = -10:10;
u = (n >= 0);
u_shift = (n >= 4);
x = 3*u - 2*u_shift;
stem(n, x, 'filled');
title('x(n) = 3u(n) - 2u(n-4)');
xlabel('n'); ylabel('Amplitude');
grid on;
```

c) Even and odd part

```
clc;
clear;
close all;
n = -10:10;
x = n.^2;   % example signal
x_even = (x + fliplr(x)) / 2;
x_odd  = (x - fliplr(x)) / 2;
```

```
subplot(3,1,1); stem(n,x); title('Original');
subplot(3,1,2); stem(n,x_even); title('Even Part');
subplot(3,1,3); stem(n,x_odd); title('Odd Part');
```



EXPERIMENT-3

AIM: DTFT of $a^n u(n)$

CODE

```
clc;
clear;
syms n w a
u = heaviside(n);
x = a^n * u;
X = symsum(x * exp(-1j*w*n), n, 0, inf);
X = simplify(X);
disp('DTFT of a^n u(n):');
disp(X);
```

Command Window

```
DTFT of a^n u(n):
piecewise(a*exp(-w*1i) == 1, Inf + 1/#E^(w*1i)*a, a*exp(-w*1i) ~=

>>
```

B) DTFT of Unit Step

CODE

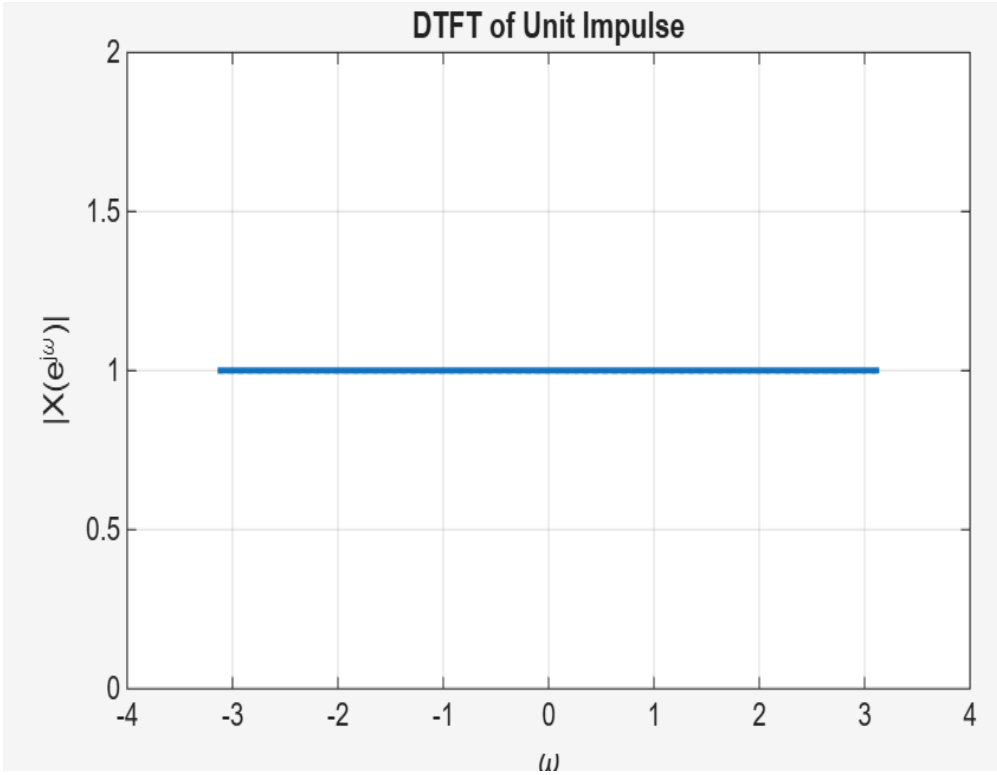
```
clc;
clear;
syms n w
u = heaviside(n);
X = symsum(u * exp(-1j*w*n), n, 0, inf);
X = simplify(X);
disp('DTFT of Unit Step:');
disp(X);
```

Command Window

```
DTFT of Unit Step:
piecewise(imag(w) < 0 & ~in(w/(2*pi), 'integer'), 1/#E^(w*1i) + e

>>
```

c) DTFT of Unit Impulse

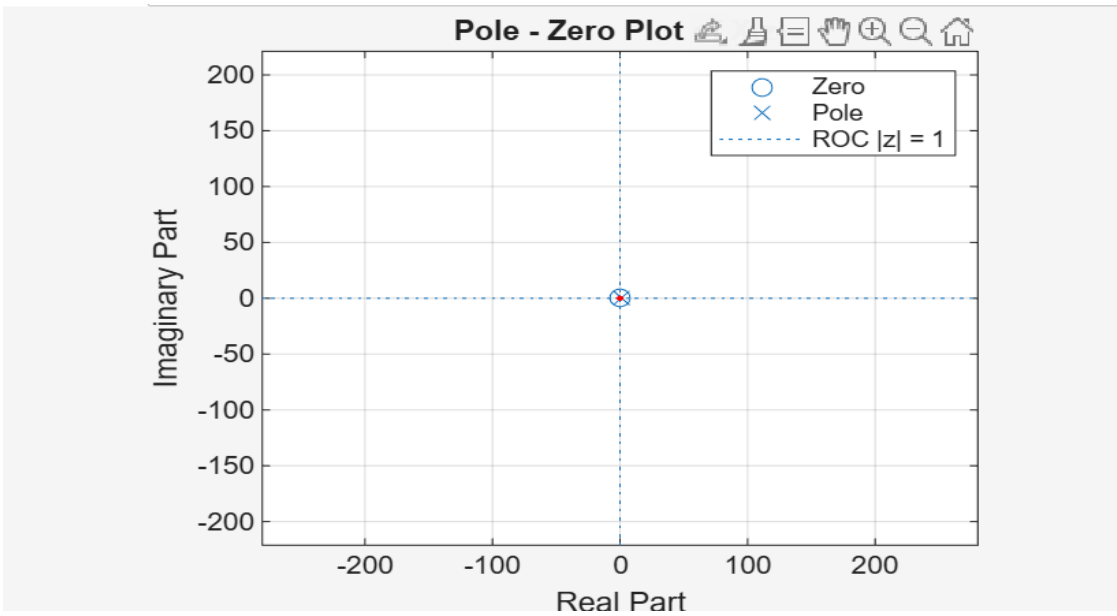
```
clc;
clear;
close all;

n = -10:10;
x = (n == 0);

w = -pi:0.01:pi;
X = zeros(size(w));

for k = 1:length(w)
   X(k) = sum(x .* exp(-1j*w(k)*n));
end

plot(w, abs(X), 'LineWidth', 2);
grid on;
xlabel('\omega');
ylabel('|X(e^{j\omega})|');
title('DTFT of Unit Impulse');
```


DTFT of Unit Impulse
|X(e^{jω})|
2
1.5
1
0.5
0
-4
-3
-2
-1
0
1
2
3
4
(ω)

EXPERIMENT-4

AIM: Z-transform of x(n)=u(n)x(n) = u(n)x(n)=u(n) and Pole-Zero Plot

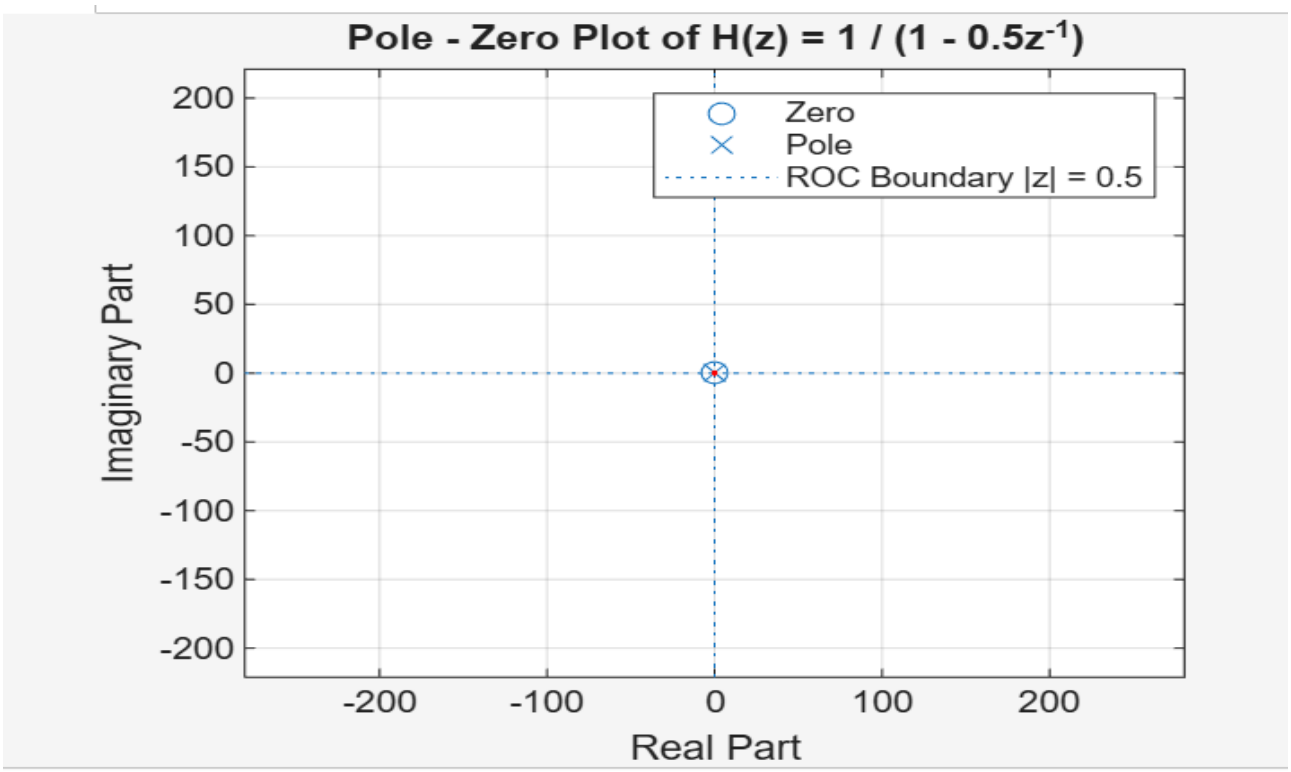
```
clc;
clear;
syms n z
% Z-transform
X = symsum(z^(-n), n, 0, inf);
disp('Z-transform X(z) = ');
simplify(X)
% Coefficients
b = [1 0];
a = [1 -1];
% Plot
figure;
zplane(b, a);
title('Pole - Zero Plot of X(z)');
grid on;
hold on;
% ROC (unit circle)
theta = linspace(0, 2*pi, 200);
r = 1;
plot(r*cos(theta), r*sin(theta), 'r--', 'LineWidth', 1.5);
legend('Zero', 'Pole', 'ROC |z| = 1');
axis equal;
```



AIM:Pole-Zero Plot of H(z)=11−0.5z−1H(z) = \frac{1}{1 - 0.5z^{-1}}H(z)=1−0.5z−11

```
clc;
clear;
close all;
% Coefficients
b = [1];
a = [1 -0.5];
% Plot
figure;
zplane(b, a);
title('Pole - Zero Plot of H(z) = 1 / (1 - 0.5z^{-1})');
hold on;
% ROC boundary (|z| = 0.5)
theta = linspace(0, 2*pi, 200);
r = 0.5;
x = r*cos(theta);
y = r*sin(theta);
plot(x, y, 'r--', 'LineWidth', 1.5);
legend('Zero', 'Pole', 'ROC Boundary |z| = 0.5');
```

```
grid on;
axis equal;
```



EXPERIMENT- 5

AIM: To Find and Plot the DFT of the following

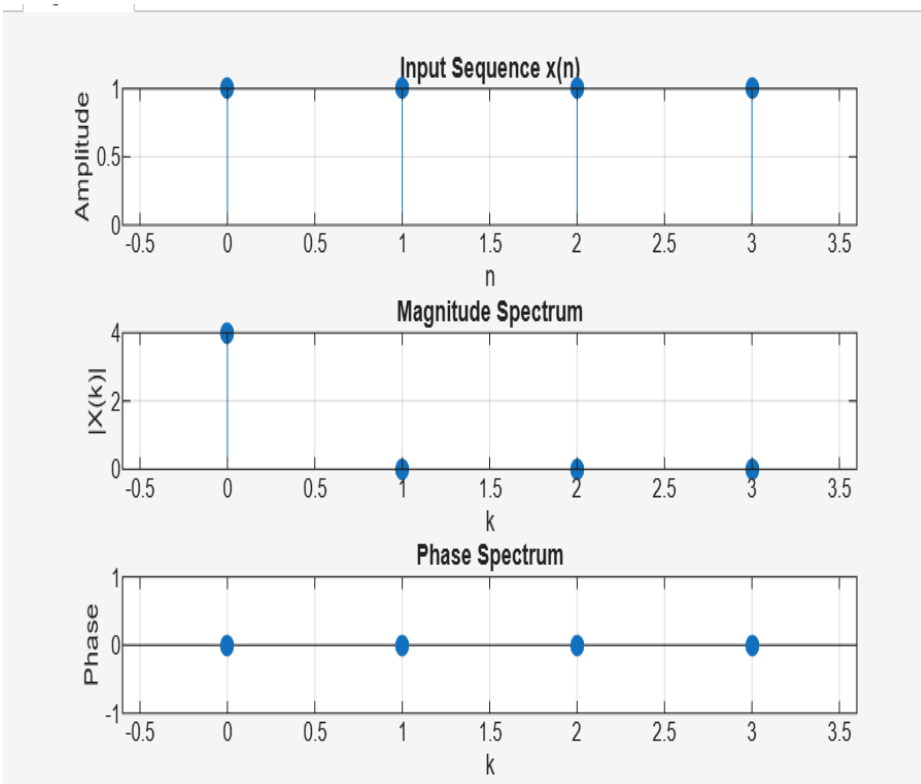
x[n]=[1 1 1 1]

Code

```
clc;
clear;
close all;
% Input sequence
x = [1 1 1 1];
N = length(x);
% DFT
X = fft(x,N);     % same as manual DFT
% Display result
disp('DFT Output:')
disp(X)
% Frequency index
k = 0:N-1;
% Plot input signal
figure;
subplot(3,1,1)
stem(0:N-1,x,'filled')
title('Input Sequence x(n)')
xlabel('n')
ylabel('Amplitude')
grid on
% Magnitude Spectrum
subplot(3,1,2)
stem(k,abs(X),'filled')
title('Magnitude Spectrum')
xlabel('k')
ylabel('|X(k)|')
grid on
% Phase Spectrum
subplot(3,1,3)
stem(k,angle(X),'filled')
title('Phase Spectrum')
xlabel('k')
ylabel('Phase')
grid on
```

Command Window

```
DFT Output:
     4     0     0     0

>>
```

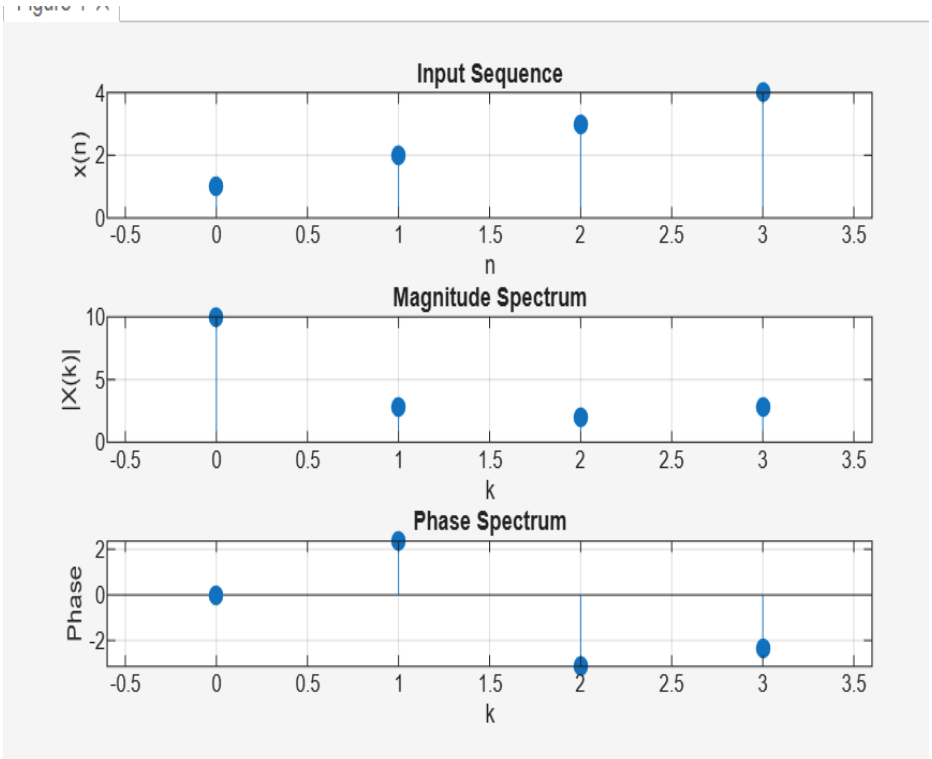


b) X[n]= [1 2 3 4]

Code

```
clc
clear
close all
% Input sequence
x=[1 2 3 4];
N=4;
% -------- DFT Calculation --------
for k=0:N-1
   X(k+1)=0;
   for n=0:N-1
```

```
    X(k+1)=X(k+1)+x(n+1)*exp(-j*2*pi*k*n/N);
  end
end
% Display Output
disp('4-point DFT is:')
disp(X)
% -------- Graphs --------
k=0:N-1;
figure
% Input sequence
subplot(3,1,1)
stem(0:N-1,x,'filled')
title('Input Sequence')
xlabel('n')
ylabel('x(n)')
grid on
% Magnitude spectrum
subplot(3,1,2)
stem(k,abs(X),'filled')
title('Magnitude Spectrum')
xlabel('k')
ylabel('|X(k)|')
grid on
% Phase spectrum
subplot(3,1,3)
stem(k,angle(X),'filled')
title('Phase Spectrum')
xlabel('k')
ylabel('Phase')
grid on
```



Command Window

```
4-point DFT is:
  10.0000 + 0.0000i  -2.0000 + 2.0000i  -2.0000 - 0.0000i  -2.000

>>
```

EXPERIMENT-6

AIM- DIT-FFT of X[n]=[0 1 2 3]

Code

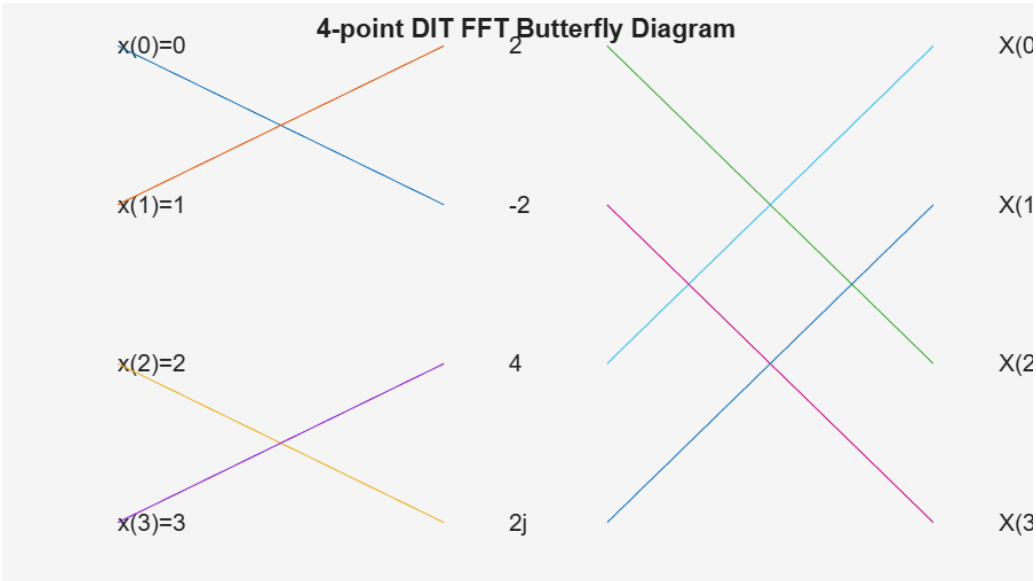
```
clc;
clear;
close all;

% Input sequence
x=[0 1 2 3];

disp('Input Sequence:')
disp(x)

%% -------- Stage 1 (Butterflies) --------
a0 = x(1)+x(3);      % 0+2
a1 = x(1)-x(3);      % 0-2

a2 = x(2)+x(4);      % 1+3
a3 = x(2)-x(4);      % 1-3

% Twiddle multiplication
W4 = exp(-1j*pi/2);
a3 = a3*W4;

%% -------- Stage 2 --------
X0 = a0+a2;
X2 = a0-a2;
X1 = a1+a3;
X3 = a1-a3;

X=[X0 X1 X2 X3];

disp('FFT Output:')
disp(X)
```

```
%% -------- Butterfly Flow Plot --------
figure
hold on
axis off

% Input labels
text(0,4,'x(0)=0')
text(0,3,'x(1)=1')
text(0,2,'x(2)=2')
text(0,1,'x(3)=3')

% First stage butterfly lines
plot([0 1],[4 3])
plot([0 1],[3 4])

plot([0 1],[2 1])
plot([0 1],[1 2])

% Stage 1 values
text(1.2,4,'2')
text(1.2,3,'-2')
text(1.2,2,'4')
text(1.2,1,'2j')
% Second stage butterfly lines
plot([1.5 2.5],[4 2])
plot([1.5 2.5],[2 4])
plot([1.5 2.5],[3 1])
plot([1.5 2.5],[1 3])
% Output labels
text(2.7,4,'X(0)=6')
text(2.7,3,'X(1)=-2+2j')
text(2.7,2,'X(2)=-2')
```

text(2.7,1,'X(3)=-2-2j')

Command Window

```
4-point DFT is:
  10.0000 + 0.0000i   -2.0000 + 2.0000i   -2.0000 - 0.0000i   -2.0000

>>
```

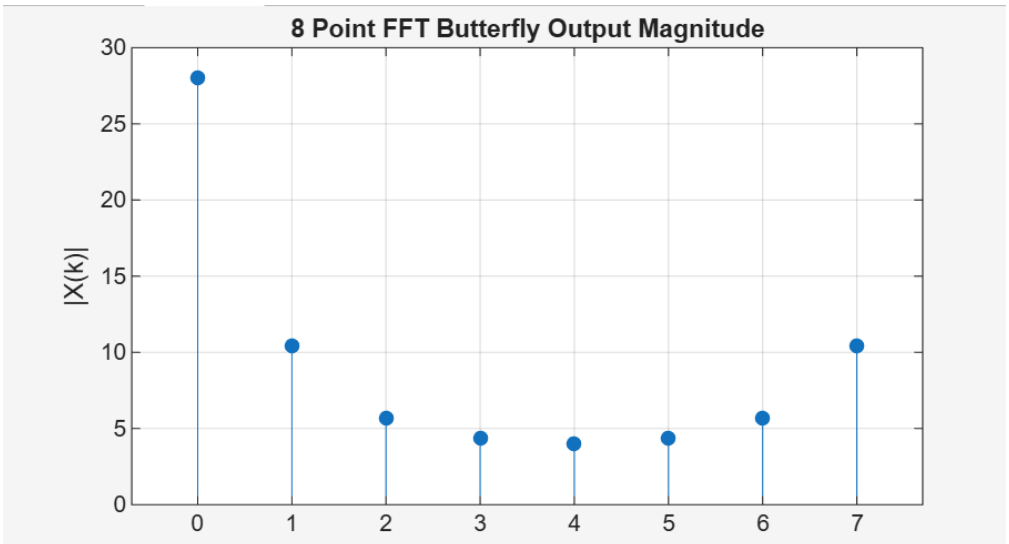


b) x[n]=[0 1 2 3 4 5 6 7]

Code

```
clc
clear
close all
x=[0 1 2 3 4 5 6 7];
disp('Input Sequence')
disp(x)
%% -------- Stage 1 --------
a0=x(1)+x(5);
a1=x(1)-x(5);
a2=x(2)+x(6);
a3=x(2)-x(6);
a4=x(3)+x(7);
a5=x(3)-x(7);
a6=x(4)+x(8);
```

```
a7=x(4)-x(8);
disp('Stage 1 Outputs')
disp([a0 a1 a2 a3 a4 a5 a6 a7])
%% Twiddle Factors
W1=exp(-1i*2*pi/8);
W2=exp(-1i*4*pi/8);
W3=exp(-1i*6*pi/8);
a3=a3*W2;
a5=a5*W1;
a7=a7*W3;
%% -------- Stage 2 --------
b0=a0+a4;
b4=a0-a4;
b1=a1+a5;
b5=a1-a5;
b2=a2+a6;
b6=a2-a6;
b3=a3+a7;
b7=a3-a7;
disp('Stage 2 Outputs')
disp([b0 b1 b2 b3 b4 b5 b6 b7])
%% -------- Final Stage --------
X0=b0+b2;
X4=b0-b2;
X2=b4-1i*b6;
X6=b4+1i*b6;
X1=b1+b3;
X5=b1-b3;
X3=b5-1i*b7;
X7=b5+1i*b7;
X=[X0 X1 X2 X3 X4 X5 X6 X7];
disp('8-point FFT Output')
disp(X)
%% Butterfly-style Output Graph
figure
```

```
stem(0:7,abs(X),'filled')
title('8 Point FFT Butterfly Output Magnitude')
xlabel('k')
ylabel('|X(k)|')
grid on
```

```
FFT Output:
  28.0000 + 0.0000i  -4.0000 + 9.6569i  -4.0000 + 4.0000i  -4.00

>>
```